

REFERENCES

- [1] Atkison, K.E., An Introduction to Numerical Analysis. John Wiley & Sons, 1978.
- [2] Birkhoff, G and Rota, G., Ordinary Differential Equations. Blaisdell, Waltman, Mass, 1962.
- [3] Burden, R.L., Faires, J.D., and Reynolds, A.C., Numerical Analysis. Prindle, Weber & Schmidt, Boston, 1979.
- [4] Fröberg, C.E., Introduction to Numerical Analysis. Addison-Wesley, 1972.
- [5] Gear, C.W., Numerical Initial-Value Problems in Ordinary Differential Equations. Prentice Hall, Englewood Cliffs, 1971.
- [6] Hamming, R.W., Numerical Methods for Scientists and Engineers. McGraw-Hill, 1962.
- [7] Isaacson, E. and Keller, H.B., Analysis of Numerical Methods. John Wiley & Sons, New York, 1966.
- [8] Kaplan, W., Advanced Calculus. Addison-Wesley, 1973.
- [9] Martin, B., Particular solutions of Difference Equations, Int. J.Math. Edu.Sci.Technol, 25,(1994), 471-473.
- [10] Rabenstein, A.L., Introduction to Ordinary Differential Equations. Academic Press, 1973.

ANSWERS TO SELECTED EXERCISES

EXERCISE 1.1

1. 11.64, 81.98, 48.36, 67.50, 4.500, 39.63.
2. 3.14, 3.142, 3.1416 with absolute errors less than $\frac{1}{2} \times 10^{-2}$, $\frac{1}{2} \times 10^{-3}$, $\frac{1}{2} \times 10^{-4}$ respectively.
3. 7.5, 6.34, 7.473, 76300, 15.24, 15.24, 122380.
4. 2.732, 35.7, 430000.
5. 73.7, 180, 541500000.
6. 0.02, 0.005, -0.0003, 40, -0.005, 0.005, -3, 0.00035, -0.029, 50, 0.0453, 0.273, 499.
7. 0.72×10^{-4} or 0.0072%.
8. a) 8633000, -426. b) 3.14159, 0.0000026.
c) 8.52, 0.005. d) 1.68, -0.005.
9. Beam measurement.

EXERCISE 1.2

1. 1143.8, which is uncertain by one unit in its last figure.
2. 0.010567. 3. 897 ± 4 (correct to only two digits).
4. a) 3.627, 3.627 ± 0.025 , 4. b) 11.15, 11.15 ± 0.02 , 11.
c) 0.0245, 0.0245 ± 0.0003 , 0.02.
5. 571. 6. $\frac{1}{9} \times 10^{-3}$, at least 3 figures.
7. 6804.0. 8. 1.773. 9. 7.1.
10. a) 0.631, 0.002. b) 287.1, 2.6.

EXERCISE 3

1. 0, 4, 9; $(2, 3, 1)^T$, $(-2, 1, 1)^T$, $(1, 0, -1)^T$.
-2, 4, 10; $(2, 3, 1)^T$, $(-2, 1, 1)^T$, $(1, 0, -1)^T$.
2, -4, -4; $(1, 3/2, 1/2)^T$; $(2, -3, 1)^T$.
2. a) 3.236; $(0.667, 1, 0.745)^T$. b) -3.414; $(-0.707, 1, -0.707)^T$.
3. 5.2361, $(0.4472, 0.8944, 1)^T$; 3, $(1, -0.5, 0)^T$;
0.7639, $(-0.4472, -0.8944, 1)^T$.
7. -1, 3, 8.

EXERCISE 4

1. b) 0.3986604. 2. 2.50987, correct to its last digit.
 3. $P = 177.4$.
 5. $P_2(x) = -3x^2 + \frac{7}{2}x$. 6. $P_2(x) = 2x^2 + 4x - 5$.

EXERCISE 5

3. 0.5580520 4. $S_n = n(n+1)(2n+1)/6$ 5. $f(x) = -0.2x^2 + 3x + 5.2$.

EXERCISE 6.1

1. a) 1.327. b) 0.921. c) 1.763.
 2. 1.365. 3. 2.92. 4. 1.320. 5. 1.3787.
 6. a) (5.6, 5.7), 5.621875. b) (51, 52), 51.59375.
 c) (3.5, 3.6), 3.584375.
 7. 1.732.

EXERCISE 6.2

1. a) 1.532. b) 0.6191 c) 2.071. 2. 4.307.
 3. 5.848. 4. 0.739085. 5. a) 0.91001 b) 1.96887.

EXERCISE 6.3

1. 4.730. 2. 0.73909. 3. a) 0.91001. b) 1.96887.
 4. i) 2.071, ii) 5.848, iii) 2.506.

EXERCISE 6.4

1. a) 5.848. b) 2.071. c) 0.7836. d) 1.065. 2. -10.261.
 3. a) 0.56714. b) -1.93375. c) 0.60710. d) 2.88324.
 e) 3.78928. f) 6.16473. g) 0.15368.
 4. 1.2. 5. 0.739085. 7. 1.723. 8. 3.478508.

EXERCISE 6.5

1. a) (51, 52), 51.5875, b) (3.5, 3.6), 3.5813
 2. 0.50
 3. $\left(\frac{e^x}{3}\right)^{1/2}$ on $[0, 1]$; 0.910.

4. 1.171. 5. 9.97.
 6. 0.347296, 1.532089, -1.879385. 7. 1.497300.
 9. 3.7893 (correct to 5 significant figures).

EXERCISE 7

1. a) $y_n = a_1 5^{n/2} \cos(n \tan^{-1}(-2)) + a_2 5^{5/2} \sin(n \tan^{-1}(-2)).$
 b) $y_n = a_1 (-1)^n + a_2 2^n.$ c) $y_n = a 4^n - 5/3.$
 d) $y_n = a_1 2^n + a_2 n 2^n + n^2 + 4n + 4.$
 e) $y_n = a_1 + a_2 (-1)^n + a_3 2^n - 2n.$
 f) $y_n = a_1 \cos \frac{n\pi}{2} + a_2 \sin \frac{n\pi}{2} + \frac{1}{2} \sin 2n + \frac{1}{2} (\cot 4 - \operatorname{cosec} 4) \cos 2n.$
 g) $y_n = a_1 \cos \frac{n\pi}{2} + a_2 \sin \frac{n\pi}{2} + \frac{1}{2} \cos n + \frac{1}{2} (\operatorname{cosec} 2 - \cot 2) \sin n.$
 h) $y_n = a_1 \left(\frac{1 + \sqrt{5}}{2} \right)^n + a_2 \left(\frac{1 - \sqrt{5}}{2} \right)^n + \frac{1}{5} 3^n.$
 i) $y_n = a_1 \cos \frac{n\pi}{2} + a_2 \sin \frac{n\pi}{2} - \frac{n}{2} \sin \frac{n\pi}{2}.$
2. a) $y_n = -\frac{2}{3} + 2^{n-1} + \frac{1}{6} 4^n$ b) $y_n = -1 + 2^n + n 2^{n-1}.$
 c) $y_n = 5 \times 2^{n-2} + \frac{53}{27} (-2)^{n-2} - \frac{1}{3} n^2 - \frac{4}{9} n - \frac{20}{27}.$
 d) $y_n = 3^{n+1} - 2$ e) $y_n = 3(1-n)2^{n-1} + n^2 2^{n-1}.$
3. $y_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1 + \sqrt{5}}{2} \right)^n + \left(\frac{1 - \sqrt{5}}{2} \right)^n \right].$
4. $I_0 = \frac{2\pi}{3\sqrt{3}}, I_1 = \frac{\pi}{3\sqrt{3}}, I_2 = \frac{-\pi}{3\sqrt{3}} + 1, I_3 = -\frac{2\pi}{3\sqrt{3}} + \frac{3}{2}, I_4 = \frac{-\pi}{3\sqrt{3}} + \frac{5}{6}.$
6. $I_0 = \frac{a^2}{4} \pi, I_1 = \frac{a^3}{3}, I_n = \begin{cases} \frac{n-1}{n+2} \frac{n-3}{n} \frac{n-5}{n-2} \dots \frac{2}{5} a^{n-1} I_1, & n \text{ is odd} \\ \frac{n-1}{n+2} \frac{n-3}{n} \frac{n-5}{n-2} \dots \frac{1}{4} a^n I_0, & n \text{ is even.} \end{cases}$
7. i) $I_{n+1} - \frac{n}{n+1} I_{n-1} = 0, I_0 = \pi/2, I_1 = 1.$

$$I_n = \begin{cases} \frac{n-1}{n} \frac{n-3}{n-2} \frac{n-5}{n-4} \dots \frac{2}{3}, & n \text{ is odd} \\ \frac{n-1}{n} \frac{n-3}{n-2} \frac{n-5}{n-4} \dots \frac{1}{2} \frac{\pi}{2}, & n \text{ is even.} \end{cases}$$

$$\text{ii) } I_{n+1} - \frac{n}{n+1} I_{n-1} = 0, I_0 = \pi/2, I_1 = 1.$$

$$I_n = \begin{cases} \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdot \frac{n-5}{n-4} \cdots \frac{2}{3}, & n \text{ is odd} \\ \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdot \frac{n-5}{n-4} \cdots \frac{1}{2} \cdot \frac{\pi}{2}, & n \text{ is even.} \end{cases}$$

$$\text{iii) } I_{n+1} + I_{n-1} = \frac{1}{n}, I_0 = \pi/4, I_1 = \frac{1}{2} \ln 2.$$

EXERCISE 8.1

1. $x_1 = 4, x_2 = 1, x_3 = 2.$
2. $x_1 = 5, x_2 = 1, x_3 = -2.$
3. $x_1 = 0.8072, x_2 = 0.2372, x_3 = -0.1046, x_4 = -0.3582.$
4. $x_1 = 0.96710, x_2 = 0.12480, x_3 = 0.42630, x_4 = 0.56790.$
5. $x_1 = 3, x_2 = 2, x_3 = 1.$
6. a) $x_1 = x_4 = -1, x_2 = x_3 = 1.$ b) $x_1 = 126, x_2 = -20, x_3 = 12.$

EXERCISE 8.2

1. $x_1 = 11, x_2 = 25, x_3 = 3.$
2. $x_1 = x_2 = x_3 = 1.$
3. $x_1 = \frac{69}{37}, x_2 = \frac{29}{37}, x_3 = \frac{10}{37}.$
4. $x_1 = -1.69, x_2 = -0.84, x_3 = 0.95.$

EXERCISE 9.1

1. $x_1 = 1.90923, x_2 = 3.19495, x_3 = 5.04479.$
2. a) 1.0137, 2.0195, 2.9961. b) 1.0001, 1.9998, -.9998, 0.9998.
3. a) 1.000, 1.000, 1.000. b) $x_1 = x_2 = x_3 = 1.000.$
- c) $x_1 = 1.104, x_2 = 2.996, x_3 = -1.021, x_4 = -2.626.$
- d) $x = 2.425, y = 3.573, z = 1.926.$
4. $x = 0.5835, y = -0.4307, z = 0.0181, t = -0.0044.$

EXERCISE 9.2

1. $x_1 = 2.0220, x_2 = 0.9899, x_3 = 3.9977.$
The exact answer is $x_1 = 2, x_2 = 1, x_3 = 4.$
2. a) $x_1 = 6.1656, x_2 = 6.0191, x_3 = -1.0892.$
- b) $x_1 = 0.8119, x_2 = 0.5638, x_3 = 5.7390, x_4 = -1.1630.$

3. $x_1 = 1.00, x_2 = 2.00, x_3 = 3.00, x_4 = -1.00.$
4. $x_1 = 1.0000, x_2 = 1.0000, x_3 = 1.0000.$
5. a) $x_1 = 0.37, x_2 = 0.04, x_3 = 0.79.$
 b) $x_1 = 0.37, x_2 = 0.04, x_3 = 0.79.$
6. $x_1 = 1.0000, x_2 = 2.0000, x_3 = -1.0000, x_4 = 1.0000.$
 $x_1 = 1.000, x_2 = 1.000, x_3 = 1.000.$
7. $x = 0.0615, y = 0.1962.$

EXERCISE 9.3

1. $x_1 = 3.001, x_2 = 2.000, x_3 = 1.000.$
2. 3.0343, 3.9714, -5.0072; 3.0037, 4.0029, -5.0057.
3. $x^{(3)} = (0.8272, 0.7565, 0.7557)^T; x^{(3)} = (1.0035, 0.9993, 0.9993)^T;$
 Exact sol: $x = (1, 1, 1)^T.$
5. $x_1 = 1.784, x_2 = 0.126, x_3 = -0.132;$ residuals are 0.001, -0.002, -0.001. and $\sum R^2 = 0.6 \times 10^{-6}.$

EXERCISE 9.4

1. 404.01. 2. a) 12321. b) 40004. c) 120004.

EXERCISE 11.1

1. Actual error = 2; error bound = $\frac{\pi^3}{2} = 2.583...$
2. Trapezoidal rule gives 0.785398, with an error bound of $6.5 \times 10^{-1},$ but no actual error.
 Simpson's 1/3 rule gives the same result with an error bound of 0.0266.
3. Trapezoidal rule gives 0.023208, with an error of 0.01160.
 Simpson's rule gives 0.032296, with an error of 0.00252.
4. Trapezoidal Rule: 0.3466, 0.0397.
 Simpson's 1/3 Rule: 0.3858, 0.0005.
 Simpson's 3/8 Rule: 0.3861, 0.0002.
5. a) 1.4972, b) 1.4775.
7. b) $I_1 = 0.7037,$ error = -3.7%
 $I_2 = 0.6855,$ error = -1.88%
 $I^* = 0.6709,$ error = -0.42%

8. Trapezoidal rule: 0.75. Simpson's $1/3$ rule: 0.69445.
Simpson's $3/8$ rule: 0.69375.

EXERCISE 11.2

1. $m = 25$, $h = .04$. 2. 1.29532.
3. $h = 1$, 1.65988841; $h = \frac{1}{2}$, 1.63751735; $h = \frac{1}{4}$, 1.63321154. The extrapolated value is 1.63177627.
4. Integral: 0.3793.
Simpson's rule gives 0.3796.
5. 0.953571. 6. 0.9163.
7. Exact value: 0.302425.
- a) 0.302506. b) 0.302425, yes.
8. a) 0.2792223. b) 0.1744644.
9. Correct value: 0.32149; Approximation: 0.32147;
Actual error: 0.00002.
10. 0.32149. 11. 0.000016. 12. 0.00000001.
13. $h \approx \sqrt{3}/100$. 14. $h = 0.14$.
15. Trapezoidal Rule: $\sqrt{3}/10^4$; Simpson's Rule: 0.014.
16. Trapezoidal rule: $h_1 = 0.4$; 0.456847, error = 1.3%
 $h_2 = 0.2$; 0.452353, error = -0.33%
extrapolated value: 0.450855, error = -0.00089%
Simpson's rule: $h_1 = 0.4$; 0.450914, error = -0.014%
 $h_2 = 0.2$; 0.450855, error = -0.00089%
extrapolated value: 0.450851, error = 0%

EXERCISE 11.3

1. 0.76692, actual error: -0.00012. 2. 0.69315.
3. a) 0.102459819, 0.102459821, 0.102459822,
The actual value is 0.102459822.
b) 1.477536117, 1.477528858, 1.477523049,
The actual value is 1.477523046.
4. 0.1183197, 0.1093104, 0.1093404, 0.1093643.

EXERCISE 12.1

1. $y = e^{\sin x}$

2. $y = x^2(e^x - e)$

EXERCISE 12.2

1. Selected values:

(a)

(b)

n	x_n	y_n App. value	y_n App. value
2	0.2	0.20010	0.9025
4	0.4	0.40360	0.81550
6	0.6	0.62264	0.74009
8	0.8	0.87939	0.67744
10	1.0	1.20318	0.62888

2. Selected values:

With $h = .1$

n	x_n	y_n App. value	$y(x_n)$ True value	$y(x_n) - y_n$ Error
2	0.2	1.01	1.01873	0.00873
4	0.4	1.05610	1.07032	0.01422
6	0.6	1.13144	1.14881	0.01737
8	0.8	1.23047	1.24933	0.01886
10	1.0	1.34868	1.36788	0.01920

With $h = .2$

n	x_n	y_n App. value	$y(x_n)$ True value	$y(x_n) - y_n$ Error
1	0.2	1.0	1.01873	0.01873
3	0.6	1.112	1.14881	0.03681
5	1.0	1.32768	1.36788	0.04020

3. Selected values:

n	x_n	y_n App. value	$y(x_n)$ True value	$y(x_n) - y_n$ Error
2	1.10	-0.906690	-0.909091	-0.002401
4	1.20	-0.836102	-0.833333	0.002769
6	1.30	-0.782055	-0.769231	0.012824
8	1.40	-0.740502	-0.714286	0.026216
10	1.50	-0.708706	-0.666667	0.042039

4. Selected values:

n	x_n	y_n
1	0.2	2.4
3	0.6	3.712
5	1.0	5.9533

5. Selected values:

n	x_n	y_n
1	0.1	0.000
5	0.5	0.030
10	1.0	0.293

6. Selected values:

n	x_n	y_n App. value	$y(x_n)$ True value	$y(x_n) - y_n$ Error
1	0.1	1.1	1.1103	0.0103
5	0.5	1.7210	1.7974	0.0764
10	1.0	3.1874	3.4366	0.2492

7. Selected values:

(a)

n	x_n	y_n App. value	$y(x_n)$ True value	$y(x_n) - y_n$ Error
2	0.2	0.42503	0.44696	0.02193
6	0.6	0.53571	0.58177	0.04606
10	1.0	0.93236	0.99126	0.05890

(b)

n	x_n	y_n App. value	$y(x_n)$ True value	$y(x_n) - y_n$ Error
1	0.2	0.45	0.44696	-0.00304
3	0.6	0.48289	0.58177	0.09888
5	1.0	0.68336	0.99126	0.30790

8. Selected values:

a)

n	x_n	y_n App. value	$y(x_n)$ True value	$y(x_n) - y_n$ Error
2	0.2	- 2.87889	- 2.98022	-0.10133
6	0.6	- 5.88039	- 6.48674	-0.60635
10	1.0	-11.56798	-13.59141	-2.02343

b)

n	x_n	y_n App. value	$y(x_n)$ True value	$y(x_n) - y_n$ Error
1	0.2	- 2.8	- 2.98022	-0.18022
3	0.6	- 5.42658	- 6.48674	-1.06016
5	1.0	-10.16756	-13.59141	-3.42385

9. Selected values:

n	x_n	y_n App. value	$y(x_n)$ True value	$y(x_n) - y_n$ Error
2	1.2	2.83	2.85619	0.02619
6	1.6	2.79432	2.84643	0.05211
10	2.0	3.04603	3.10364	0.05761

EXERCISE 12.3

1. a) $y_1 = 0.4261$, $y_2 = 0.5042$, $y_3 = 0.5933$.

b) $y_1 = 0.4263$, $y_2 = 0.5045$, $y_3 = 0.5937$.

2.

n	x_n	$y(x_n)$ True value	(a)		(b)	
			y_n App. value	$y(x_n) - y_n$ Error	y_n App. value	$y(x_n) - y_n$ Error
1	0.1	1.1103	1.115	-0.0047	1.1105	-0.0002
2	0.2	1.2428	1.2476	-0.0048	1.2432	-0.0004

3.

n	x_n	(a)	(b)
		y_n	y_n
1	0.1	-0.8955	-0.8953
2	0.2	-0.7821	-0.7816

4.

n	x_n	(a)	(b)
		y_n	y_n
1	0.1	-2.4394	-2.4438
2	0.2	-2.9730	-2.9835
3	0.3	-3.6181	-3.6370

EXERCISE 12.4

1.

n	x_n	y_n App. value	$y(x_n)$ True value	$y(x_n) - y_n$ Error
1	0.1	0.9512711	0.9512711	0.0
2	0.2	0.9051709	0.9051709	0.0
3	0.3	0.8618342	0.8618343	0.0000001

2.

n	x_n	y_n
1	1.1	-0.0103307
2	1.2	-0.0425944
3	1.3	-0.0984319
4	1.4	-0.1788454

3.

n	x_n	y_n
1	1.1	0.34591028
2	1.2	0.86662168
3	1.3	1.60718134
4	1.4	2.62031130

4.

a)

n	x_n	y_n
1	0.2	0.710453334
2	0.4	0.609525248

b)

n	x_n	y_n
1	0.1	1.004837500
2	0.2	1.018730901
3	0.3	1.040818422
4	0.4	1.070320289
5	0.5	1.106530934

c)

n	x_n	y_n
1	0.1	0.095162500
2	0.2	0.181269098
3	0.3	0.259181577

5.

a)

n	x_n	y_n App. value	$y(x_n)$ True value	$y(x_n) - y_n$ Error
1	0.1	0.09033286	0.09033301	0.00000015
2	0.2	0.16265642	0.16265669	0.00000027

b)

n	x_n	y_n App. value	$y(x_n)$ True value	$y(x_n) - y_n$ Error
1	0.1	1.1055127	1.1055127	0.0
2	0.2	1.2242082	1.0242084	-0.1999998

c)

n	x_n	y_n App. value	$y(x_n)$ True value	$y(x_n) - y_n$ Error
1	0.1	1.000333385	1.0003334	0.000000015
2	0.2	1.002670217	1.0026702	-0.000000017
3	0.3	1.009040609	1.0090406	-0.000000009

EXERCISE 12.5

1. $y_5 = 0.35181$.
2. a) $y_4 = 0.9896$, $y_5 = 0.9794$. b) $y_4 = 0.9909$, $y_5 = 0.9796$.
3. a) $y_4 = 0.389257987$, $y_5 = 0.479225518$.
b) $y_4 = 0.389231882$, $y_5 = 0.479219705$.
4. With starting values obtained by Euler's method:
a) $y_4 = 0.28157103$, $y_5 = 0.32265044$.
b) $y_4 = 0.283015994$, $y_5 = 0.321273249$.

EXERCISE 13

1. c) $x_1 = 1.31722$, $y_1 = 2.15000$, $x_2 = 1.67828$, $y_2 = 2.40134$, $h = 0.1$.
d) $x_1 = 0.21495$, $y_1 = 0.11450$; $h = 0.1$.
2. $x = 1.01880$, $y = 2.02109$.
- 5.

a)

n	x_n	y_n App. value	$y(x_n)$ True value	$y(x_n) - y_n$ Error
1	1.2	1.9083	1.9067	-0.0016
2	1.4	1.9904	1.9886	-0.0018
3	1.6	2.2115	2.2100	-0.0015
4	1.8	2.5520	2.5511	-0.0009

b)

n	x_n	y_n App. value	$y(x_n)$ True value	$y(x_n) - y_n$ Error
1	0.2	0.3600	0.3611	0.0011
2	0.4	0.6449	0.6465	0.0016
3	0.6	0.8531	0.8547	0.0016
4	0.8	0.9765	0.9776	0.0011